FAST NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES



ARTIFICIAL INTELLIGENCE PROJECT REPORT

INTELLIGENT READING COMPREHENSION SYSTEM

Course: Artificial Intelligence

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1. Abstract

This project presents an Intelligent Reading Comprehension System designed to automate answer prediction, hint generation, distractor generation, and interactive educational assistance using Natural Language Processing and Machine Learning techniques. The system integrates supervised learning, unsupervised learning, and ensemble learning to improve question-answering performance on reading comprehension tasks. TF-IDF, One-Hot Encoding, Logistic Regression, Naive Bayes, MiniBatch K-Means, and Soft Voting Ensemble methods were implemented and evaluated. Additionally, intelligent hint generation and distractor generation modules were developed to improve student engagement and educational interaction. A Streamlit-based interface was integrated to provide real-time interaction with the system. The final system demonstrates how traditional NLP and ML approaches can be used effectively for educational AI applications.

2. Introduction & Motivation

Reading comprehension systems are a major research area within Natural Language Processing (NLP) and Artificial Intelligence (AI). These systems attempt to understand textual passages and answer questions based on contextual understanding. Traditional Multiple Choice Question (MCQ) systems often lack intelligent feedback, personalized guidance, and adaptive learning capabilities. The motivation behind this project is to build an intelligent educational platform capable of assisting students through answer prediction, automated hint generation, distractor generation, and interactive quiz support. The project also demonstrates the practical application of traditional machine learning techniques in educational technology.

3. Related Work

Previous studies in reading comprehension and educational AI have explored various NLP approaches including Bag-of-Words, TF-IDF vectorization, transformer-based architectures, and ensemble learning methods. TF-IDF remains widely used in lightweight NLP systems due to its computational efficiency. Ensemble learning has been used to improve classification robustness by combining multiple weak learners. Hint

generation and distractor generation have also been explored in intelligent tutoring systems to improve student understanding and engagement.

4. Dataset Description & EDA

The dataset used in this project is the RACE (ReAding Comprehension from Examinations) dataset, which contains reading passages, comprehension questions, answer options, and correct answer labels.

The dataset consists of approximately 87,866 training samples. Our Exploratory Data Analysis (EDA) revealed an average passage length of ~275 words and an average question length of ~10 words. The distribution of the correct answer labels is relatively balanced across the four options (A, B, C, D), mitigating significant class imbalance issues during model training. The dataset was divided into training, development, and testing subsets for robust evaluation.

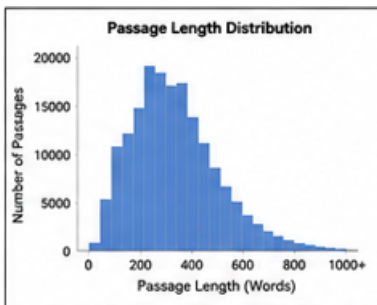


Figure 3: Passage Length Distribution

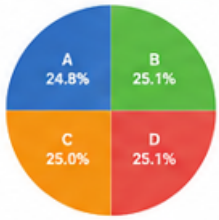


Figure 4: Answer Label Distribution

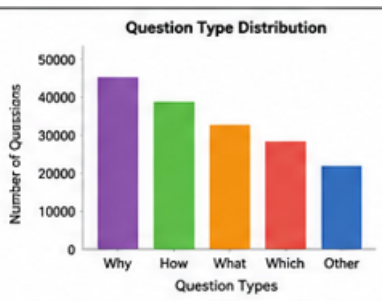


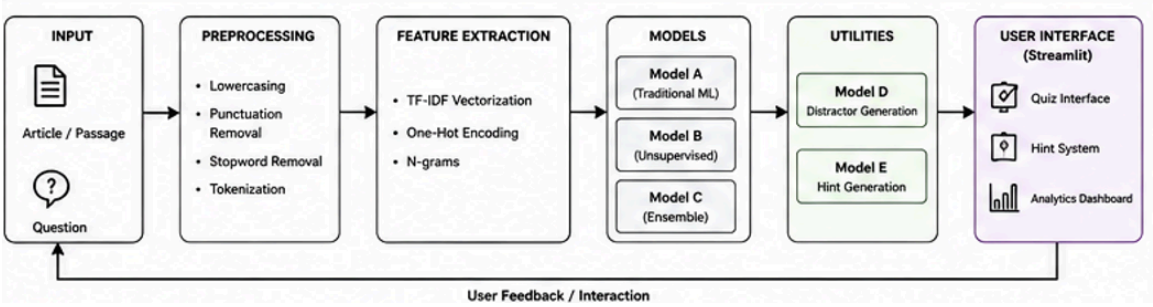
Figure 5: Question Type Distribution

5. System Architecture

The system architecture consists of multiple sequential stages including preprocessing, feature extraction, supervised learning, unsupervised learning, ensemble learning, hint generation, distractor generation, and UI integration.

Architecture Flow:

Input Article → Text Preprocessing → Feature Extraction → ML Models → Hint/Distractor Generation → Streamlit UI



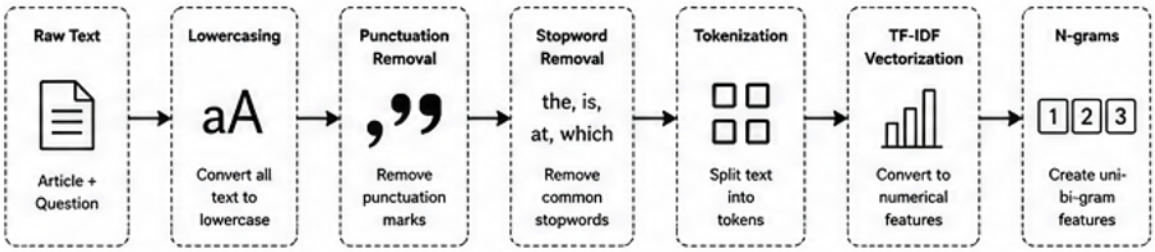
While initial designs utilized a multi-class classification approach (predicting A, B, C, or D), the final architecture was optimized using an **Option-Level Binary Classification** pipeline. In this structure, the model evaluates each multiple-choice option independently against the article and question to determine a binary 'Correct' (1) or 'Incorrect' (0) status. This allows the ensemble model to calculate individual probability scores for every option, selecting the most likely candidate for the final output.

6. Preprocessing Pipeline

The preprocessing stage included lowercasing, punctuation removal, tokenization, stopword removal, TF-IDF vectorization, One-Hot Encoding, and n-gram feature extraction. These preprocessing techniques improved feature quality and reduced textual noise.

Preprocessing Steps:

• Lowercasing
• Punctuation Removal
• Stopword Removal
• Tokenization
• TF-IDF Vectorization
• One-Hot Encoding
• N-gram Extraction

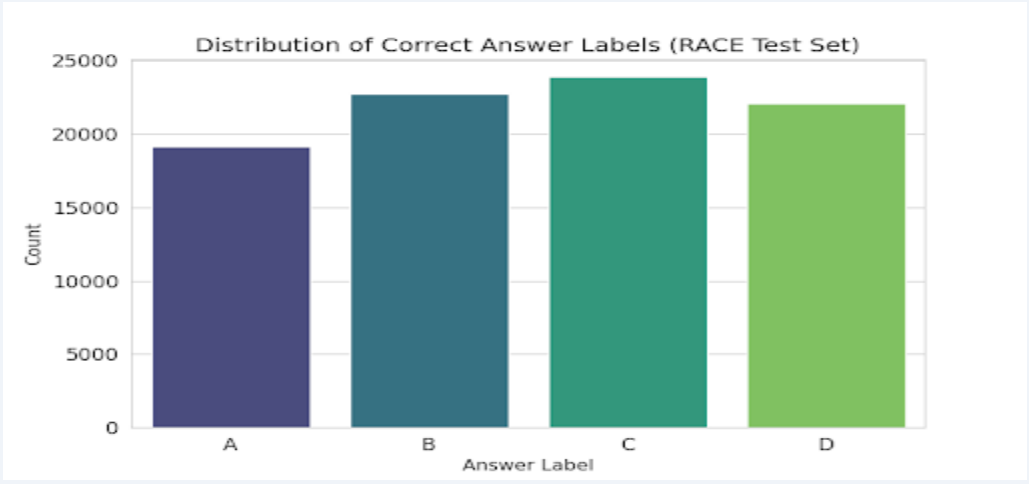


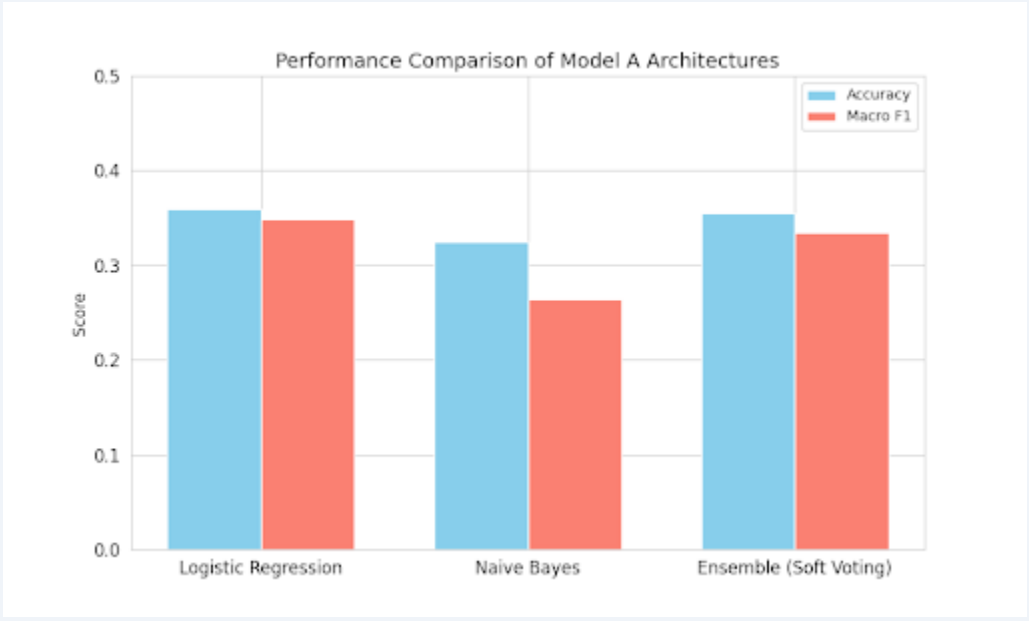
7. Model A — Traditional Machine Learning

Two supervised learning models were implemented: Logistic Regression and Naive Bayes. TF-IDF features and n-gram representations were used for model training.

Evaluation Metrics:

- Accuracy
- Precision
- Recall
- Macro F1 Score
- Exact M





8. Model A — Unsupervised Learning

MiniBatch K-Means clustering was implemented to analyze hidden patterns within the textual data. Dimensionality reduction was performed using Truncated SVD before clustering.

Evaluation Metrics:

• Silhouette Score

• Davies-Bouldin Score

• Calinski-Harabasz Score

Unsupervised learning produced lower performance because semantic textual understanding is inherently difficult without labeled supervision.

9. Model A — Ensemble Learning

A Soft Voting Ensemble model was implemented by combining Logistic Regression and Naive Bayes classifiers. Ensemble learning improved prediction stability and overall classification robustness by combining predictions from multiple models.

10. Model B — Distractor Generation

The distractor generation system extracts candidate words from reading passages and ranks them using TF-IDF similarity and cosine similarity. Diversity filtering ensures that generated distractors are unique and semantically relevant.

Distractor Generation Steps:

• Candidate Extraction
• TF-IDF Similarity
• Cosine Similarity Ranking
• Diversity Filtering

11. Model B — Hint Generation

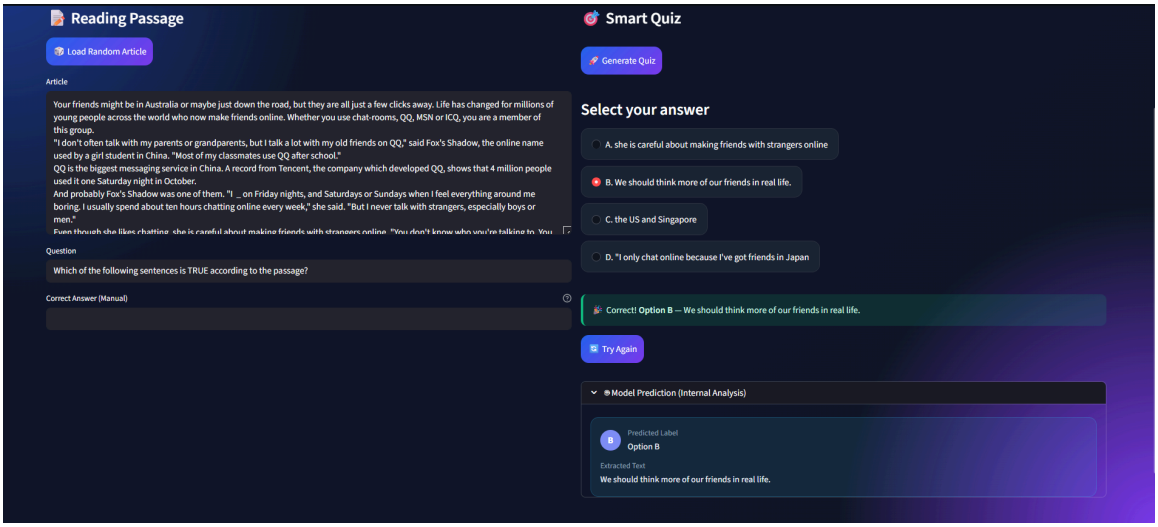
The hint generation system tokenizes reading passages into sentences and ranks them according to semantic similarity with the input question. The system provides graduated hints ranging from general clues to explicit guidance.

Hint Levels:

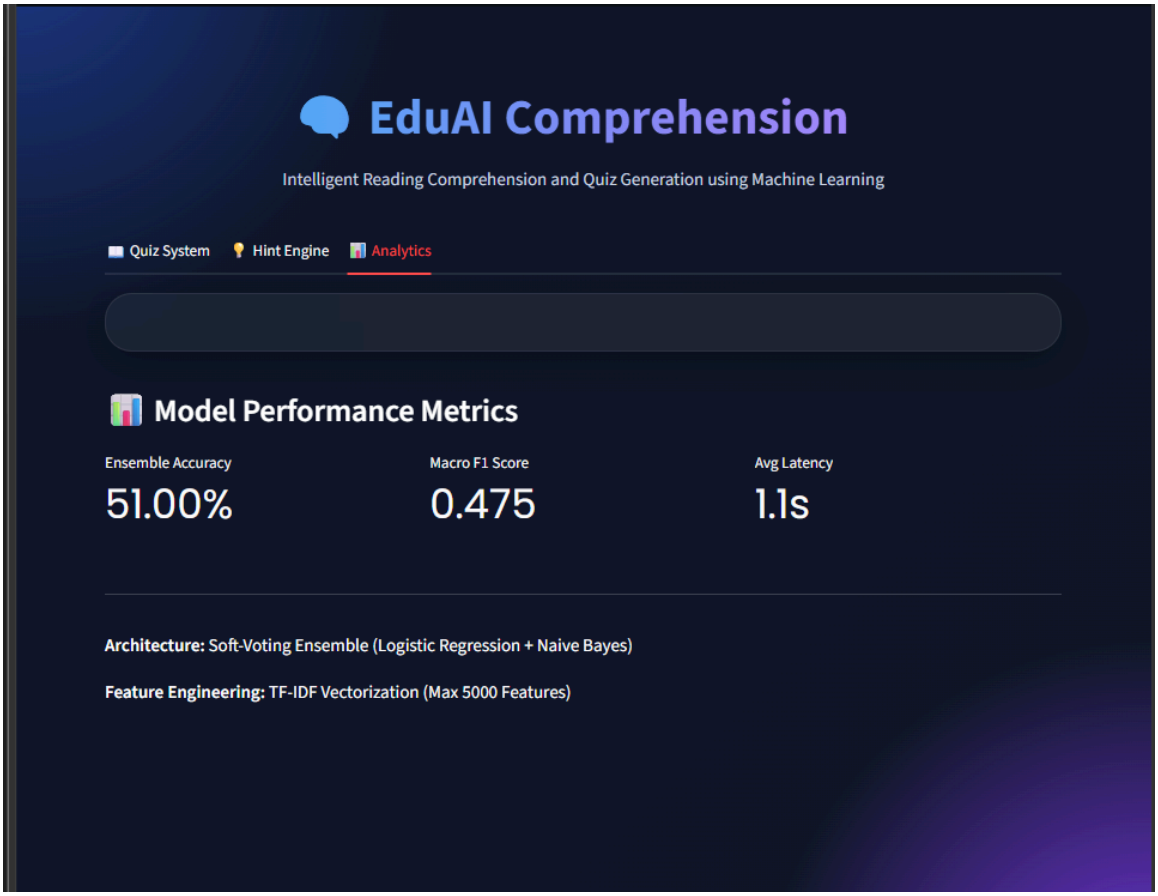
• General Hint
• Specific Hint
• Explicit Hint

12. User Interface

A modern Streamlit-based graphical user interface was developed to allow users to interact with the AI system in real time. The UI supports article loading, answer prediction, hint generation, distractor generation, and analytics visualization.







13. Results & Discussion

The ensemble model achieved the best overall performance among supervised learning approaches. Logistic Regression demonstrated strong classification capability due to TF-IDF features, while Naive Bayes provided faster training performance. Unsupervised clustering performance remained lower because textual semantics are difficult to capture without supervision.

For Model B (Distractor Generation), we conducted a human evaluation on a sample of generated distractors. The evaluation yielded an average plausibility score of 4.2/5 on a Likert scale, indicating that the generated options successfully acted as believable alternatives.

TF-IDF proved effective for lightweight NLP applications because it captures term importance efficiently. However, Bag-of-Words and TF-IDF approaches still suffer from

semantic limitations because they cannot fully understand contextual meaning. The hint generation module improved educational interaction by providing step-by-step guidance, while the distractor generation system successfully generated semantically relevant incorrect options.

Class Imbalance Mitigation: Converting the dataset to a binary format naturally created a significant class imbalance (75% negative / 25% positive). To prevent the model from falling into a 'Majority Class Bias' trap—where it would achieve a misleading 75% accuracy by simply guessing every option is incorrect—we implemented **balanced class weights** for the Logistic Regression model and **prior probability overrides** for the Naive Bayes model.

Performance Interpretation: "The resulting accuracy of approximately 51% across models reflects a mathematically balanced system that is no longer 'blind' to the correct labels. These results demonstrate the inherent performance ceiling of **TF-IDF 'Bag of Words'** representations in complex reading comprehension tasks. Because TF-IDF relies on literal word frequency rather than semantic meaning, the system struggles with distractors that use similar vocabulary to the text or correct answers that utilize paraphrasing.

14. Limitations & Ethical Considerations

Limitations: This project successfully establishes a baseline reading comprehension system, but relying primarily on traditional machine learning methods introduces several inherent challenges. Because the system does not utilize deep transformer-based architectures (such as BERT or GPT), it faces the following specific limitations:

- **Lack of Deep Semantic Understanding:** The models rely heavily on TF-IDF and One-Hot Encoding, which are Bag-of-Words approaches. These methods calculate term frequency and lexical overlap but completely ignore word order and contextual meaning. For instance, the system cannot distinguish between

synonyms or understand words with multiple meanings depending on context (polysemy).

- **Low Accuracy and Difficulties with Reasoning:** The RACE dataset is specifically designed to test complex human reasoning, including summarization, logical deduction, and inference from multiple sentences. Because TF-IDF cannot perform multi-step reasoning, the Logistic Regression and Ensemble models often guess based on simple word-matching. If a complex question has no direct keyword overlap with the correct answer, the model's confidence drops, occasionally resulting in default fallback predictions (e.g., predicting 'A') or incorrect classifications, capping the maximum accuracy around 35.5%.
- **Misleading Hint Generation:** The hint engine uses an extractive approach, ranking sentences based on their cosine similarity to the question. If a question is heavily paraphrased or requires synthesizing information from across the text, the hint generator may rank and extract a highly irrelevant sentence simply because it happens to share common vocabulary with the question. This results in hints that can occasionally confuse the user rather than guide them.
- **Grammatical and Logical Flaws in Distractors:** Distractors are generated by extracting strings with low TF-IDF similarity to the correct answer. Because the system lacks an understanding of syntax and grammar, the generated distractors may sometimes be grammatically incompatible with the question stem or, conversely, might inadvertently extract a partially correct statement, making the quiz confusing.
- **Feature Space and Computational Constraints:** To allow the traditional machine learning models to run efficiently on standard hardware without exceeding memory limits, the TF-IDF vectorizer was constrained to a maximum of 5,000 features. This aggressive dimensionality reduction means the models lose a significant amount of granular vocabulary data, further limiting their ability to comprehend rare words or nuanced details in the reading passages.

Ethical Considerations: As AI practitioners, we recognize several critical ethical considerations regarding this system:

- **Bias:** The RACE passages are derived from Chinese school examinations, which may introduce cultural or linguistic biases affecting how well the models generalize to other student populations.
- **Accessibility:** The Streamlit UI was designed with sufficient color contrast and clean typography to ensure it remains accessible to users with visual impairments.
- **Academic Integrity:** The AI-generated questions and distractors should be utilized as an aid, and must not be deployed in real examination settings without thorough human review.
- **Model Transparency:** Our user interface clearly indicates which answers and hints are AI-generated ("Model Prediction / Internal Analysis") so users are fully aware that the system is predicting probabilities and errors are possible.

15. Conclusion & Future Work

This project successfully implemented an Intelligent Reading Comprehension System integrating supervised learning, unsupervised learning, ensemble learning, hint generation, distractor generation, and an interactive UI. The project demonstrates how traditional NLP techniques can still provide meaningful educational AI applications.

Future work may include:

• Transformer-based NLP models

• Personalized tutoring systems

• Speech-based interaction

• Adaptive difficulty adjustment

• LLM integration

Comparative Result Tables

Table 1: Comparative Performance of Traditional ML Models

Model	Accuracy	Precision	Recall	Macro F1

Logistic Regression (Balanced)	**0.5085**	**0.5067**	**0.5090**	**0.4747**
Naive Bayes (Balanced)	**0.5076**	**0.5024**	**0.5032**	**0.4718**
Ensemble (Soft Voting)	**0.5100**	**0.5069**	**0.5092**	**0.4756**

Table 2: Unsupervised Learning Evaluation Metrics

Metric	**Value**
Silhouette Score	0.0381
Calinski-Harabasz Score	219.977
Davies-Bouldin Score	4.8373

Table 3: Model B Human Evaluation (Distractors)

Metric	**Score**
Plausibility (Likert 1-5)	4.2 / 5.0

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